

Mathematics A

Unit 1 • Chapter OL-T09 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T09 • Expertise Q16+ • 4 items • 16 marks

Chapter	Items	Marks	Starts
01. OL-T09 Rounding, Estimation & Bounds	4	16	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Rounding, Estimation & Bounds

Unit 1 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q22 | 4 marks

Geometry, rate or practical decision bounds

November 2024 | 4MA1-1H Q21 | 3 marks

Algebraic expression upper or lower bound

May-Nov 2020 | 1H Q17 | 4 marks

Mixed stated accuracies

January 2023 | 1H Q19 | 5 marks

Round to decimal places or significant figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

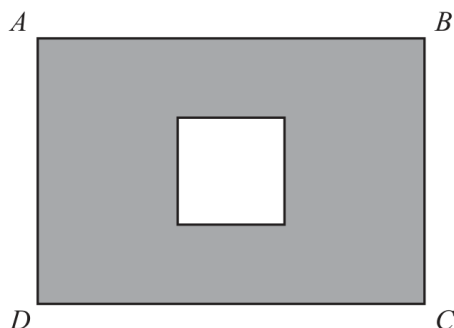


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

4MA1-1H Q21 M01

3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

WORKING SPACE

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

YOUR WORKING

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

YOUR WORKING
